This is the first mathematical law discovered for earthly bodies: the law of falling bodies.[fn4] Up until this point, only mathematical laws for the movements of the planets had been discovered. Mathematical perfection is no longer confined to the heavens.

But the greatest result is still to come, and it will be Isaac Newton who will attain it. Newton studies in depth the results of Galileo and Kepler and by combining them, finds the hidden diamond. We can follow his reasoning in terms of the 'little moon', as he himself tells it in *The Mathematical Principles of Natural Philosophy*, the book that forms the foundation of modern science.

Imagine that the Earth, writes Newton, had many moons, like Jupiter. In addition to the real Moon, imagine other moons, and in particular a *little moon* which orbits the Earth at a minimal distance from it, just above the mountain peaks. At what speed would this little moon travel? One of the laws discovered by Kepler relates the radius of orbit with period, namely, the time taken to complete an orbit.[fn5] We know the radius of the orbit of the real Moon (Hipparchus had measured it in antiquity) and its period (one month). We know the radius of the orbit of the little moon (the radius of the Earth, measured by Eratosthenes in antiquity). With a simple proportion we can compute the period of the orbit of the little moon. The result is one hour and a half. The little moon would complete its orbit around the Earth every one hour and a half.

Now, an object which orbits does not go straight: it continually changes direction, and a change of direction is an *acceleration*. The little moon accelerates towards the centre of the Earth. This acceleration is easy to compute.[fn6] Newton makes the simple calculation, and the result is … 9.8 metres per second per second! The same acceleration as in Galileo's experiments for falling bodies on Earth.

Coincidence? It can't be, reasons Newton. If the *effect* is the same – a downwards acceleration of 9.8 metres per second per second – the *cause* must be the same. And so: *the force which causes the little moon to turn around its orbit must be the same as that which causes objects to fall to the ground on Earth*.

We call the force causing objects to fall gravity. Newton understands that it is this same gravity that makes the little moon turn around the Earth. Without this gravity it would run away in a straight line. But then, also, the real Moon must orbit the Earth because of gravity! And the moons which orbit Jupiter are attracted by Jupiter, and the planets which turn around the Sun are attracted by the Sun! Without this attraction, every celestial body would move in a straight line. So the universe, then, is a large space where bodies attract